

Structure of Book

Title

The Paradigm of Paradox

Structure

Promotional Verbiage

Title Page

Dedication

Preface

Acknowledgements

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Start each chapter with a Paradigm Discourse

Use a quote

Start every chapter with a self-referential comic.

Use threads

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The Realm

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9-Dots

QT3

3D Chess

Red herring puzzles

Ignoring conjecture

Infinitely thin line paradigm

Three Act play

Paradigm – Protagonist

Dogma – antagonist

Understanding

The objective of science is to understand, more than to just predict

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This should be added.

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Argument Outline

Part 1

1. Paradigms
2. Paradox
3. Relativity
4. Uncertainty
5. Abstraction
6. Interference
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8. Loops
9. QTP

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18. Logical Bases
19. Formal Systems

Argument (Elevator Pitch)

The universe is nonlocal, thank you John Bell, but it comes with a cosmic speed limit, the speed of light, thank you Einstein. These are not compatible, yet the universe does not break. Asimov observed that the beginning of discovery is not "Eureka" (I've found it) but rather, "that's odd." Odd it is that quantum correlations require something to break the speed of light, yet whatever that something is, it seems that it cannot be used to send an FTL signal, it seems that it cannot be explained by a spacelike cause. But Gisin has correctly observed that some new *third kind* of

causality is required to explain nonlocal correlations and spacelike causality is an obvious candidate.

The hypothesis presented here tackles this conundrum with a speculative idea based on several advances in disparate fields, so before presenting the formal hypothesis, we must introduce a couple of concepts, one in relativity, the other in nonlinear logic.

A special spacelike interval can be inferred from an invariant in relativity, called a symmetric spacetime interval (SSI), which prescribes a situation dependent preferred reference frame from which all observers can agree on which of two correlated but spacelike separated events is the cause and which the effect. The SSI allows us to break the symmetry of space and distinguish left from right, which for spacelike intervals are preserved quantities, making spacelike causality realistically consistent. For timelike intervals it is the earlier and later events which are the invariants. This capitalizes on the symmetry between space and time.

A pair of such spacelike causes, put in opposite Poincaré rotations, can be used to send information into the past light cone of the later cause. This raises the specter of temporal paradox, the single most effective argument against spacelike causes, FTL, time travel, the works. Thus, spacelike causality presents itself as a logical impossibility even before it is a physical impossibility. Surely, this negates any possible viable theory of spacelike causality.

However, in 1969, G. Spencer-Brown introduced imaginary truthvalues (i, j) as the solution to logical paradox, and later work showed that they form a conjugate basis set with the Boolean truthvalues (T, F). Simple self-referential logical forms are multi-valued. They will be paradoxical in one basis, but indeterminate in another. The Liar's Paradox is paradoxical in the Boolean basis, but indeterminate in the imaginary basis, while the Circular Argument is paradoxical in the imaginary basis but indeterminate in the Boolean basis.

This suggests a way to overcome the paradox objection. Imagine two spacelike causes with opposite Poincaré rotations that are connected on both ends with timelike causes. On a spacetime diagram they look like a 'bowtie', a closed causal loop in spacetime. Coincidentally, and suggestively, this loop looks a lot like the sign for infinity. Note that this loop is inherently *nonlocal*. Now imagine that if both spacelike legs are represented in one basis, the causality is paradoxical, but if represented in the conjugate basis they are merely indeterminate. Nature can avoid the paradox by the expedient act of collapsing in the indeterminate basis. The loop has measured itself.

Which basis is indeterminate depends on the details of the entanglement, but this idea supports an *objective measurement mechanism*, introduces *nonlinearity* into quantum mechanics naturally through the agency of *self-reference*, determines the *measurement basis*, automatically generates *randomness*, and specifies how an isolated quantum system can *self-collapse* by removing the *artificial distinction* between a measurement system and a quantum system. It explains *nonlocality* as a form of *spacelike causality*, in a *relativistically consistent* way, but by collapsing in the *indeterminate basis* acts as a censor mechanism to *prevent temporal paradox*. It resolves most of the *interpretation* issues and promises *new technologies* with which to manipulate nature. That is a tempting explanatory list.

This hypothesis is referred to as Quantum Temporal Paradox (QTP), which perhaps should be more accurately referred to as quantum temporal indeterminacy, but that's a less catchy name.

QTP puts limits on spacelike causality, but this should not surprise us. Physics puts limits on timelike causality too, forward only, conservation laws, etc. The purpose of this book is to characterize those limits. It is a quest; there is no guarantee that a theory of spacelike causes with the required properties can be formulated. The fact that the measurement mechanism has gone unsolved for a hundred years implies a nearly intractable paradigm barrier. Somewhere along the line, we have accepted hidden assumptions from which there is no solution to the problem; they need to be rooted out. One way or another, such a quest will deepen our understanding of quantum mechanics.

To pull this off requires a couple of additional breakthroughs.

Measurements of entangled quantum systems cannot be used for FTL because the randomness hides any differences in the choice of measurement bases. Attempts to correlate randomness by using two receiving channels led to the discovery of entanglement swapping and GHZ states. Eberhard's proof that spacelike causality by quantum measurement cannot get past the randomness issue continues to stand.

In 2006 Quantum Tic-Tac-Toes was published which hinted at two things; first, that semi-entanglements, two-particle quantum systems that are halfway between separable and an EPR state, could be chained together locally to defeat the randomness objection. Alice and Bob are spacelike separated observers who receive a pair 3 particle semi-entanglements. The sender can connect these in two ways, locally, changing the correlations between the particles observed by the receiver, thus sending an FTL signal.

Second that causality is nonlocal, being distributed both in time and in space. The classical view is that time progresses linearly, that the present is an infinitely thin separator between past and future. However, if causality is distributed nonlocally, then the present becomes a window around 'now' where the past, present, and future, become ambiguous. Talking about causality within this expanded temporal window from a classical perspective has one using the language of time travel.

There remain two challenges that have so far resisted analysis. It is not clear if there are linear operators which can cross entangle a pair of semi-entanglements via local operations. Such entanglements may require nonlinear operators which will mandate changes in the foundations of quantum mechanics and until experimentally tested are the most vulnerable point of falsifiability of this hypothesis. However, given such operators, a robust theory of decoherence can be constructed.

The final challenge is showing how a closed causal loop in spacetime consisting of two spacelike causes with opposite Poincare' rotations and a timelike cause can be paradoxical in some bases,

but indeterminate in one-and-only one. If each of the spacelike causes consisted of cyclic entanglements which were then measured, by at least the receiver, then they have already collapsed to classical values. This would short-circuit the self-collapse in the indeterminate basis.

Paradigm Discourses

Inspired by GEB and even Galileo's dialogues. Major conceptual tools are promoted to characters, all recruited and mentored by Understanding. Paradigm is the protagonist, Dogma the antagonist.

List of Characters

1. Understanding
2. Dogma
3. Curiosity
4. Theory
5. Experiment
6. Logic
7. Proof
8. Mathematics
9. Reason
10. Paradigm
11. Faith
12. Epiphany

List of Discourses

Act 1

1. Boxes – **Paradigms**
 - a. *A story is worth a thousand pictures.*
2. Self-Reference – **Paradox**
 - a. *What are the odds that a random guess will get the right answer to this question?*
3. Causality – **Relativity**
 - a. *There once was a lady named Bright – Limerick*
4. Firm Color – **QM**
 - a. *Truth is not relative; it's black or white, so or not so – Logic*
5. Ignoring – **Uncertainty**
 - a. *To see or not to see, that is the question.*
6. Crest Valley – **Interference**
 - a. *Quantum mechanics: the dreams stuff is made of.*
7. Spooky-Coins and Magic-Envelopes – **Entanglement**
 - a. *Don't ask how it can be like that, no one knows. It is a black pit from which no returns. – Feynman*
8. All You Zombies – **QTP**
 - a. *Once you've eliminated the impossible...*
9. The Quest – **R&D**
 - a. *The assurance of things hoped for; the conviction of things not seen.*

Act 2

10. Journal – **Objections**
 - a. *And this too shall pass away.*
11. Outsmarted – **QT3**
 - a. *Which came first, the present or the past?*
12. Paired Correlations – **By Contradiction**
 - a. *Science is the belief in the ignorance of experts.*
13. Red, White and Blue – **12 Light Seconds**
 - a. *Hidden information.*
14. Barrier Boxes – **World Ribbons**
 - a. *The merely difficult we do right away, the impossible takes a little longer.*
15. Surf Beach – **SSI**
 - a. *Philosophy quiz, "Define force."*
16. Art Show – **Bowtie**
 - a. *What goes around comes around.*
17. Cloning – **Teleportation**
 - a. *Clones are people too.*
18. Bingo Bos – **Indistinguishable**
 - a. *Murphy's law was propounded by another man of the same name.*
19. Halfway There – **Quantum Optics**
 - a. *Five out of four people have trouble with fractions.*
20. Faster Than Light – **FTL by Indistinguishability**
 - a. *It's a rare person who wants to hear what he doesn't want to hear.*
21. There and Back – **Now and Then**
 - a. *time travel. Quantum experiment proves it*
22. Numbskull – **QTI**
 - a. *If you try to fail, and succeed; which have you done?*
23. Friends – **Critique**
 - a. *The kisses of an enemy are deceitful, but the wounds of a friend are faithful.*

Next

- NLL – Logic
- Semi-Entanglements – Mathematics
- Extending entanglements – Curiosity
- Pruning by contradiction – Proof
- SSI – Experiment
- Decoherence – Reason
- GHZ – Theory
- Cyclic entanglements - Dogma

Act 3

24. Frame – **Search Party**
 - a. *quote*
25. LOF – **In the Beginning**
 - b. *Quote*
26. Engineering – **Circle of Truth**
 - c. *quote*
27. NLL – **Introductions**

- d. *Quote*
- 28. Smullyan – **Discourse**
- e. *quote*
- 29. Tech – **Discourse**
- f. *Quote*

next

TBD

- 1. Sync Noise – **FTL by Measurement**
 - a. *An egotist is someone who is more interested in themselves than they are in me.*
- 2. Faster Than Letter – **SSI**
 - a. *It's a rare person who wants to hear what he doesn't want to hear.*
- 3. Out of Order – **Now**
 - a. *Which came first, the present or the past?*
- 4. Outsmarted – **QT3**
 - a. *The past is a different country; they do things differently there.*
- 5. Chess
- 6. Tbd
 - a. The English spelling rule, 'i' before 'e' except after 'c' disproved by science.

Character Profiles

Understanding

The wise old man, leader of the group, metaphor for the **true goal of science**. Generally, in the background, typically laze fair leadership style but is flexible on style and can switch to autocratic, bureaucratic, mentor, etc. on a dime. Host. Dresses conservatively. Modeled on Major Metzler and Dr. Joe. The constant; minimal to no character development. Loves books. Fears failing his team. He develops the other characters. What he believes in needs to cover all the others.

“What can be achieved is what one should believe in.”

“There are three reasons to believe something; it is true, when what you believe affects what is so, or because it is worth believing in.”

“Choose to believe what is true, or what can be true if you believe in it, or what is worth believing in.”

If reality informs belief, then belief informs reality.

“In classical physics, reality informs belief; in quantum physics belief informs reality.”

“In linear situations, reality informs belief. In nonlinear ones, belief may inform potential. But it is what is worth believing in that informs meaning.”

“Reality generally informs belief. In nonlinear situations, by contrast, belief has the potential to inform reality by affecting what can be. But it is what is worth believing in that informs meaning.”

Philosophy – “What advances comprehension can be believed.”

Paradigm

Protagonist. Most recent member. Metaphor for **thinking out-of-the box**. Searches for the pattern, passionate about understanding (grok) reality. Teases Dogma. Accepts point on the QTP quest. Playful, low tolerance for stupidity, but patient in explaining. Consistently sees patterns where others see noise. Loves to claim he prefers words that start with ‘p’.

Mathematics school – **a new unnamed one**, driven by self-reference.
 Philosophy – “What has explanatory power can be believed.”

Dogma

Antagonist. Metaphor for the **stodgy establishment**. Has been with Understanding for a long time. Likes coverage; knowledge should be complete but minimal, w/o gaps or duplication, structured for rapid retrieval, utilization and understanding. Originally phrased as “Likes a ‘place for everything and everything in its place’.” Initially nemesis to Paradigm. Detail oriented. Stressed by change. Keeps a journal. Very slowly comes around to Paradigm.

Philosophy – “What can be rigorized can be believed.”

Curiosity

Female. Metaphor for the **thirst for knowledge**. High energy and inquisitive; many but not too many of her lines are questions. Next to Understanding has the highest EQ in the group. Has a crush on Paradigm she denies. Spent 19 months in the hospital as a child recovering from some unspecified illness or accident. Classy in carriage and dress, but young at heart. Generous and forgiving.

Philosophy – “What is worth believing in can be believed.”

Philosophy – “Belief is born from surprise, ‘I don’t believe it’, is the beginning of discovery.”

Reason

Interested in synthesis. Metaphor for **valid explanations**. Interested in applications. A keen observer, he appreciates the team diversity. Wide ranging interests, a generalist. Fond of puns. Practiced with a Yo-yo. Loves pursuing Curiosity, just because it bugs her. Extrovert, a flair for the dramatic, likes practical jokes.

Mathematical school – **intuitionist**, we can trust our judgments of truth via a valid argument.

Philosophy – “When all other possibilities have been eliminated, what is left can be believed.”

Logic

Analytical. Metaphor for **formal systems**. Likes clear arguments, finds ambiguity frustrating. Fast thinker. Playful. Amateur magician loves the good-natured con, despises gullibility. Scuba diver. <need more, in particular need some sharp contrast with Proof> Should they be twins?

Mathematical school – **logisticism**, all of math can be derived from logic.

Philosophy – “There is only one justification to believe anything – because it is true.”

Proof

Sceptic & agnostic. Female? Metaphor for **Gödel like limits**. Likes step-by-step development. Slow thinker, but sassy retorter. Likes jigsaw puzzles. He/She is the last hold-out to Paradigm, when everyone else expected Dogma to be. Compulsive about finishing a task.

Mathematical school – **formalism**, axioms, symbolic derivation of theorems.

“When the chain of causation is unbroken, it can be believed.”

“Beliefs are only secure when the chain of causation is unbroken.”

“Beliefs are only secure when there’s an unbroken chain of causation.”

Philosophy – “What can be argued can be believed.”

Mathematics

Mature. Metaphor for **quantification**. Carries around paper and pencil, but usually does the calculations in his head. Does not use a calculator but fascinated by Understanding’s new computation engines. Good at statistics, a keen observer. Likes working in wood, ‘measure twice, cut once.’ Knows all the units, will call out anyone who gets them wrong, or forgets them. Owns an irrational clock.

Mathematical school – **set theory**, foundational basis for logic and math.

Philosophy – “What can be quantified can be believed, within error bars.”

Theory

Professor type. Metaphor for **natural law** and predictability. Often assumes the professorial tone of voice. Likes abstractions, generalizations, and formulas. Art lover motivated by beauty. Major surprise - also has a law degree; can argue any side. Teams, teases, and tarries with Experiment.

Philosophy – “What is beautiful can be believed.”

Experiment

Practical type. Metaphor for **falsifiability**. Loves hands-on. Likes demonstrations. Distracted by bright baubles. Loves amusement park rides. Bold, solves problems by diving in and just trying stuff. Has a huge Lego collection. Teams, teases, and tarries with Theory.

Philosophy – “I believe what I see.”

Faith

Enigmatic. Older female. Metaphor for **wise theology**. Wise, insightful, motherly, but old west hard-headed type. The ideal shrink. Generally amused by life’s oddities. Sees the tapestry of life and can trace the threads. Scary good at predicting the future. She has hope for the future.

“What has been tested by time can be believed.”

“Believe in what can be redeemed; the redemption of ignorance, the redemption of error, the redemption of hubris.”

“If it has overcome ignorance, error and hubris, it might be believable.”

Philosophy – “Suspend belief until it has overcome ignorance, error and hubris.”

Epiphany

Female. Flighty. Classic blond. Metaphor for **unexpected insight**. Likes picking up random objects and placing them next to each other. Tends to ask dumb questions which inspire deeper inquiry or insight. Will randomly redirect a conversation. Tactile. Dresses flamboyantly but pulls it off. Unpredictable in nearly every area. Modeled on cosmetic girl in borrowed car which ran out of gas because there was no *idiot* light.

Philosophy – “Trust the wisdom of the crowd; what others believe can be believed.”

Lists of Character Attributes

Hate the duplication but it is important for internal consistency. Above are the character profiles, a few paragraphs. Below are the lists that confirm attributes are not duplicated. These lists ensure that characters are different and help to define the tensions between them.

Character Metaphors

1. Understanding – **true goal of science**
2. Paradigm – **thinking out-of-the box**
3. Dogma – **stodgy establishment**
4. Curiosity – **thirst for knowledge**
5. Reason – **valid explanations**
6. Logic – **formal systems**
7. Proof – **Gödel like limits**
8. Mathematics – **quantification**
9. Theory – **natural law**
10. Experiment – **falsifiability**
11. Faith – **wise theology**
12. Epiphany – **unexpected insight**

Character Philosophies

1. Understanding – “What advances comprehension can be believed.”
2. Paradigm – “What has explanatory power can be believed.”
3. Dogma – “What can be rigorized can be believed.”
4. Curiosity – “What is worth believing in can be believed.”
5. Reason – “When all other possibilities have been eliminated, what is left can be believed.”
6. Logic – “There is only one justification to believe anything – because it is true.”
7. Proof – “What can be argued can be believed.”
8. Mathematics – “What can be quantified can be believed, within error bars.”
9. Theory – “What is beautiful can be believed.”
10. Experiment – “I believe what I see.”
11. Faith – “Suspend belief until it has overcome ignorance, error and hubris.”
12. Epiphany – “Trust the wisdom of the crowd; what others believe can be believed.”

Character Passions

1. Understanding – pushing back the darkness
2. Paradigm – knowing *why*
3. Dogma – structure
4. Curiosity – relationships
5. Reason – eliminating error
6. Logic – aloofness
7. Proof – skepticism
8. Mathematics – calculation shortcuts
9. Theory – elegant simplicity
10. Experiment – diving in, ‘we don’t need no stinking plan.’
11. Faith – meaning
12. Epiphany – having a good time

Character Fears

1. Understanding – failing the team
2. Paradigm – being stupid
3. Dogma – change
4. Curiosity – letting people down
5. Reason – being conned
6. Logic – being alone
7. Proof – ambiguity
8. Mathematics – blunders
9. Theory – the realm
10. Experiment – the realm
11. Faith – not being true to first principles
12. Epiphany – being ostracized

Character Attributes Template

1. Understanding – tbd
2. Paradigm – tbd
3. Dogma – tbd
4. Curiosity – tbd
5. Reason – tbd
6. Logic – tbd
7. Proof – tbd
8. Mathematics – tbd
9. Theory – tbd
10. Experiment – tbd
11. Faith – tbd
12. Epiphany – tbd

Relationship Profiles

Theory-Experiment

Thick as thieves. Tease each other a lot but respect each other and tend to stick together.

Paradigm-Dogma

Protagonist-antagonist. Paradigm finds him stuffy and inflexible. Dogma finds him flighty and undisciplined. Eventually they warm to each other.

Paradigm-Curiosity

Relies on her but doesn't realize it. She is in love with him but will deny it.

Reason-Curiosity

Reason courts Curiosity, in part from a genuine affection, but mostly because it annoys her; like Herb and Jennifer (WKRP Cincinnati).

Faith-Dogma

Mother-son.

Relationship

Next

Relationship

Next

Author P-Types

Pesky
Pedagogical
Perspiring
Perplexing
Pernicious
Precocious
Pedantic
Persnickety

Intrepid Reader
Indomitable intrepid reader
Insatiable intrepid reader

Chapter Order

text

Strategy 3

Act I – In which our protagonist finds his quest.

1. Paradigms – the world view that informs/constrains our perceptions
 - a. PD – “Boxes” – Always in a box.
 - b. Quote – *A story is worth a thousand pictures.*
 - c. Comic – **Unknown**: “Wagon on square wheels” (need permission)
 - d. Text – 9 Dot Puzzle, Kuhn
2. Paradox – resolve self-reference with dual valued truthvalues in conjugate bases
 - a. PD – “Self-Reference” – Game of 13
 - b. Quote – *<random guess> - credit?*
 - c. Comic – **Foxtrot**: “reviewing book on logic” (need permission)
 - d. Text – Imaginary truthvalues
3. Relativity – NLS backstory
 - a. PD – “Causality”
 - b. Quote – *Bright limerick*
 - c. Comic – **Frank & Ernest**: “Two sticks, counter-intuitive isn’t it!”
 - d. Text – spacetime, invariance, intervals, spacelike, etc.
4. Quantum Mechanics – Enigmatic
 - a. PD – “Firm Color” – conjugate bases
 - b. Quote – *Truth is not relative, it’s black or white, either so or not so.*
 - c. Comic – **xkcd**: “Amber particles of grain” (with permission)
 - d. Text – Stupefying assault, state, superposition, Dirac, bases.
5. Uncertainty – Conjugate bases
 - a. PD – “Ignoring” – ignoring is the key to abstraction
 - b. Quote – *To see, or not to see, that is the question.*
 - c. Comic – **Wilkinson**: “Estrogen Studies” (need permission)
 - d. Text – Superposition, action, the uncertainty principle, red herrings.

6. Interference – Destructive cancellation
 - a. PD – “Crest Valley” – multiple path interference
 - b. Quote – *Quantum mechanics: the dreams stuff is made of.*
 - c. Comic – **Foxtrot**: “Why do I always feel like I’m being watched?”
 - d. Text – Interferometer, delayed choice, mix/ensembles, measurement problem.
 7. Entanglement – Non separable states
 - a. PD – “Spooky Coins & Magic Envelopes”, white – theme
 - b. Quote – Don’t ask how it can be like that, no one knows. It is a black pit from which no returns. – Feynman
 - c. Comic – **Speed Bump**: “Prisoner Plagiarism” (need permission)
 - d. Text – Separable vs non separable states, Surez & Scarini, conflict with relativity
 - e. Quantum iago?
 - f. Bell’s inequality explained?
 8. Quantum Temporal Paradox (QTP) – collapse in indeterminate basis
 - a. PD – “All You Zombies” – Dogma’s rant, all sign onto the quest
 - b. Quote – *Once you’ve eliminated the impossible...*
 - c. Comic – **Frank & Ernest**: “Global” (need permission)
 - d. Text – Summary, Heinlein.
 9. R&D – The pros are the motivation, but the science is in the cons.
 - a. PD – “The Quest” – Faith makes her appearance
 - b. Quote – *The assurance of things hoped for; the conviction of things not seen.*
 - c. Comic – **Frank & Ernest**: “Procrastination” (need permission)
 - d. Text – Story better way to communicate paradigm shift.
- Act II – In which our protagonist embarks on the quest with many twists and turns.
10. Objections – Table of initial objections
 - a. PD – “Journal” – Dogma’s journal, self-reflection, and plan
 - b. Quote – *And this too shall pass away.*
 - c. Comic – **Frank & Ernest**: – “Kleptomania clinic” (need permission)
 - d. Text – Pros, Cons, Table of Objections 1.0 (4 items).
 11. Quantum Tic-Tac-Toe – Objection #1: Temporal Paradox
 - a. PD – “Outsmarted” – Logic defeats Paradigm by attacking his past.
 - b. Quote – *The past is a foreign country; they do things differently there.*
 - c. Comic – **BC**: “Boomerang” (need permission)
 - d. Text – 125, chronblocks, attack the past, ½ pts, perfect play, interference.
 12. Lightseconds – desc
 - a. PD – “Red, White and Blue” – Dual tracks, SSI analog.
 - b. Quote – *Hdden nformaton*
 - c. Comic – **Fox Trot**: “Football Pattern” (need permission)
 - d. Text – Forest metaphor: farmer, waking up, orchard.
 13. Symmetric Spacetime Interval – Objection #2: Relativistic Consistency
 - a. PD – “Numbskull” – Dogma demos imaginary coins
 - b. Quote – *If you try to fail, and succeed, which have you done?*
 - c. Comic – **Far Side**: “My brain is full” (need permission)
 - d. Text – Thick worldlines, counterfactuals, causality stories, derivation
 14. FTL by Measurement – Objection #3: Quantum Randomness
 - e. PD – “Correlations”

- f. Quote – *An egotist is someone who is more interested in themselves than they are in me.*
 - g. Comic – **Unknown**: “Waste of Time” (need permission)
 - h. Text – Bad GHZ
15. Misconceptions – desc
- a. PD – “What are Friends For?” – Team takes Paradigm down a notch or two.
 - b. Quote – *The kisses of an enemy are deceitful, but the wounds of a friend are faithful* – *Old Testament*
 - c. Comic – **Far Side**: “Beach Boards” (need permission)
 - d. Text – Make clear the nature of the failures and inadequacies.
16. Oops – Virtually empty chapter.
- a. PD – “Missing” – Paradigm is missing, Understanding exercises leadership.
 - b. Quote –
 - a. *You always find it in the last place you look.* – *pessimist*
 - b. *Well, I sure hope you stop looking once you’ve found it.* – *wit*
 - c. Comic – **Mother Goose and Grimm**: “God Sneezed” (need permission)
 - d. Text – Oops.
17. Decoherence – desc
- a. PD – “Epiphany”
 - b. Quote – *A mind once expanded by a new idea, never regains its original shape* – *Mark Twain*
 - c. Comic – **name**: “title” (need permission)
 - d. Text – brief
18. Title – desc
- e. PD – “Title”
 - f. Quote – *italic*
 - g. Comic – **name**: “title” (need permission)
 - h. Text – brief
19. Next
20. https://lichess.org/editor/rnbqkbnr/pppp1ppp/8/4p3/3PP3/8/PPP2PPP/RNBQKBNR_w_KQkq_-_0_1
21. tbd

Strategy 4

Act I – In which our protagonist finds his quest.

- 1. Paradigms – the world view that informs/constrains our perceptions
 - a. PD – “Boxes” – Always in a box.
 - b. Quote – *A story is worth a thousand pictures.*
 - c. Comic – Wagon on square wheels carry a load of round ones (need permission)
 - d. Text – 9 Dot Puzzle, Kuhn
- 2. Paradox – resolve self-reference with dual valued truthvalues in conjugate bases
 - a. PD – “Self-Reference” – Game of 13
 - b. Quote – *<random guess> - credit?*
 - c. Comic – **Foxtrot**: “reviewing book on logic” (need permission)
 - d. Text – Imaginary truthvalues
- 3. Relativity – NLS backstory

- a. PD – “Causality”
 - b. Quote – *Bright limerick*
 - c. Comic – **Frank & Ernest**: “Two sticks, counter-intuitive isn’t it!”
 - d. Text – spacetime, invariance, intervals, spacelike, etc.
 4. Quantum Mechanics – Enigmatic
 - a. PD – “Firm Color” – conjugate bases
 - b. Quote – *Truth is not relative, it’s black or white, either so or not so.*
 - c. Comic – **xkcd**: “Amber particles of grain” (with permission)
 - d. Text – Stupefying assault, discrete, state, superposition, Dirac, bases, puns.
 5. Uncertainty – Conjugate bases
 - a. PD – “Ignoring” – ignoring is the key to abstraction
 - b. Quote – *To see, or not to see, that is the question.*
 - c. Comic – **Wilkinson**: “Estrogen Studies” (need permission)
 - d. Text – Superposition, action, the uncertainty principle, red herrings.
 6. Interference – Destructive cancellation
 - a. PD – “Crest Valley” – multiple path interference
 - b. Quote – *Quantum mechanics: the dreams stuff is made of.*
 - c. Comic – **Foxtrot**: “Why do I always feel like I’m being watched?”
 - d. Text – Interferometer, delayed choice, mix/ensembles, measurement problem.
 7. Entanglement – Non separable states
 - a. PD – “Spooky Coins & Magic Envelopes”, white – theme
 - b. Quote – *Don’t ask how it can be like that, no one knows. It is a black pit from which no returns.* – Feynman
 - c. Comic – **Speed Bump**: “Prisoner Plagiarism” (need permission)
 - d. Text – Separable vs non separable states, Surez & Scarini, conflict with relativity
 - i. Quantum iago?
 - ii. Bell’s inequality explained?
 8. Quantum Temporal Paradox (QTP) – collapse in indeterminate basis
 - a. PD – “All You Zombies” – Dogma’s rant, all sign onto the quest
 - b. Quote – *Once you’ve eliminated the impossible...*
 - c. Comic – **Frank & Ernest**: “Global” (need permission)
 - d. Text – Summary, Heinlein.
 9. R&D – The motivation is in the pros, but the science is in the cons.
 - a. PD – “The Quest” – Faith makes her appearance.
 - b. Quote – *The assurance of things hoped for; the conviction of things not seen.*
 - c. Comic – **Frank & Ernest**: “Procrastination” (need permission)
 - d. Text – Story better way to communicate paradigm shift.
- Act II – In which our protagonist embarks on the quest with many twists, turns and setbacks.
10. Objections – Table of initial objections
 - a. PD – “Journal” – Dogma’s journal, self-reflection, and plan
 - b. Quote – *And this too shall pass away.*
 - c. Comic – **Frank & Ernest**: – “Kleptomania clinic” (need permission)
 - d. Text – Pros & Cons of QTP, table of objections.
 11. QTP3 – Objection #1: Temporal Paradox
 - a. PD – “Outsmarted” – Logic defeats Paradigm by attacking his past.
 - b. Quote – *The past is a foreign country; they do things differently there.*

- c. Comic – **BC**: “Boomerang” (need permission)
 - d. Text – 1-2-5, chronoblocks, attack the past, ½ pts, perfect play, interference.
- 12. World Ribbons – Objection #2: QM Incompatible with Relativity
 - a. PD – “Barrier Boxes”
 - b. Quote – *The merely difficult we do right away, the impossible takes a little longer.*
 - c. Comic – **Ziggy**: “Hard to think outside the box” (need permission)
 - d. Text – Collapse is not *instantaneous*, it is *spacelike*. Happenings event occurrence
- 13. 12 Light Seconds – Objection #2: QM Incompatible with Relativity
 - a. PD – “Red, White, and Blue”
 - b. Quote – *Hidden information*
 - c. Comic – **Foxtrot**: “Football Pattern” (need permission)
 - d. Text – Orchard metaphor, the 12 letter invariant.
- 14. Packets – Objection #3: Relativistic Consistency
 - a. PD – “Faster Than Letter”
 - b. Quote – *It’s a rare person who wants to hear what he doesn’t want to hear.*
 - c. Comic – **name**: “desc” (need permission)
 - d. Text – tbd
- 15. Now – Objection #3: Relativistic Consistency
 - a. PD – “Out of Order”
 - b. Quote – *What came first, the present or the past?*
 - c. Comic – **name**: “desc” (need permission)
 - d. Text – tbd
- 16. Closed SSI Loops – Objection #1: Temporal Paradox
 - a. PD – “Numbskull” – Dogma demos imaginary coins
 - b. Quote – *If you try to fail, and succeed, which have you done?*
 - c. Comic – **Far Side**: “My brain is full” (need permission)
 - d. Text – tbd
- 17. FTL by Measurement – Objection #4: Quantum Randomness
 - a. PD – “Synchronized Noise”
 - b. Quote – *An egotist is someone who is more interested in themselves than they are in me.*
 - c. Comic – **unknown source**: “Waist of time” (need permission)
 - d. Text – Bad GHZ
- 18. Misconceptions – desc
 - a. PD – “What are Friends For?” – Team takes Paradigm down a notch or two.
 - b. Quote – *The kisses of an enemy are deceitful, but the wounds of a friend are faithful.*
 - c. Comic – **name**: “title” (need permission)
 - d. Text – Make clear the nature of the failures and inadequacies.
- 19. Oops – Virtually empty chapter.
 - a. PD – “Missing” – Paradigm is missing, Understanding exercises leadership.
 - b. Quote – *italic*
 - c. Comic – **name**: “title” (need permission)
 - d. Text – Oops.
- 20. Decoherence – desc
 - a. PD – “Epiphany”

- b. Quote – *A mind once expanded by a new idea, never returns to its original shape*
– Mark Twain
 - c. Comic – **name**: “title” (need permission)
 - d. Text – brief
21. Title – desc
- a. PD – “Title”
 - b. Quote – *italic*
 - c. Comic – **name**: “title” (need permission)
 - d. Text – brief
22. next

Plan for QTP Reveal

Title “All You Zombies” refers to Heinlein’s famous time travel story.

1. Possibility that entanglement supports spacelike causality.
 - a. Paradigm quotes a fictional character.
 - b. Dogma begs for a hunt for hidden variables.
 - c. Dogma salivates before the dinner bell.
2. How many kinds of causes?
 - a. Paradigm doesn’t like limits.
 - b. Dogma’s rant
 - c. Curiosity’s “on the other hand” tease
3. Closed causal loop of spacelike causes.
 - a. Temporal communication
 - b. Light cone, “Don’t know why that name.”
 - c. Simultaneity not preserved
 - d. Left/right ordering is preserved
 - e. The slender triangle, message received before being sent
 - f. Casual paradox
 - g. Reality disappears in a puff of logic.
4. Understanding’s challenge to Paradigm.
 - a.

Stats – there is a spreadsheet...

Sizes of the discourses and chapters.

Chapter	Discourse	Technical
1	1585	3430
2	849	904
3	497	2494
4	1172	2349
5	1329	1177
6	483	2248
7	1133	1524
8	2020	860
9	665	492

10	814	576
11	1897	3087
12	975	2367
13	904	1616
14	1285	2152
15	1212	1794
16	1181	
17	1353	
18	3119	
19	13	

Plans

Plan for QTP Reveal

Title “All You Zombies” refers to Heinlein’s famous time travel story.

5. Possibility that entanglement supports spacelike causality.
 - a. Paradigm quotes a fictional character.
 - b. Dogma begs for a hunt for hidden variables.
 - c. Dogma salivates before the dinner bell.
6. How many kinds of causes?
 - a. Paradigm doesn’t like limits.
 - b. Dogma’s rant
 - c. Curiosity’s “on the other hand” tease
7. Closed causal loop of spacelike causes.
 - a. Temporal communication
 - b. Light cone, “Don’t know why that name.”
 - c. Simultaneity not preserved
 - d. Left/right ordering is preserved
 - e. The slender triangle, message received before being sent
 - f. Casual paradox
 - g. Reality disappears in a puff of logic.
8. Understanding’s challenge to Paradigm.
 - a.

Plan for QT3

Someone consistently beats the inventor of the game, because they have figured out quickly that the correct strategy is to attack the past. The inventor should be Paradigm, except that it ties him too closely to me, but getting beaten makes him more vulnerable and likeable.

Curiosity previous had the interference rant. She could show interference in QT3.

Reason’s uncertainty box has the conjugate bases but is more than just superposition.

Need to use QT3 to introduce semi-entanglements.

Experiment, Proof, & Reason had the SCME EPR demo.

Mathematics and Proof work together to use Understanding’s computation engine to prove that X has a winning strategy.

Understanding has been largely background.
 Theory and Experiment appear to have no roles in this.
 Logic gets to be the young whipper snapper who beats Paradigm
 Dogma's inner dialog is the narrator.

"Well," he said, "what have you all come up with?" Understanding surveyed the group.

Plan for SSL Progression

12. World ribbons
13. 12 lsec invariant
14. Dual EPR spacelike cause
15. Poincare rotation of 0.6c
16. Temporal paradox/indeterminacy
17. Non causal signaling.

Lists

Each Paradigm Discourse is started with a quote, in italics.
 Each Chapter is started with a self-referential comic.

Quotes

In Use

1. *A story is worth a thousand pictures.*
2. *What are the odds...*
3. *There once was a lady named Bright – Limerick*
4. *Truth is not relative; it's black or white, so or not so – Logic*
5. *To see or not to see, that is the question.*
6. *Quantum mechanics; the dreams stuff is made of.*
7. *Don't ask how it can be like that, no one knows. It is a black pit from which no one returns. – Feynman*
8. *Once you've eliminated the impossible...*
9. *The assurance of things hoped for; the conviction of things not seen.*
10. *And this too shall pass away.*
11. *Which came first, the present or the past?*
12. *Science is the belief in the ignorance of experts. – Feynman*
13. *Hdden nformaton*
14. *The merely difficult we do right away, the impossible takes a little longer.*
15. *Philosophy quiz, "Define force." Brazen student, "Make me."*
16. *What goes around comes around.*
17. *Clones are people two.*
18. *Murphy's law was propounded by another man of the same name. – Nagler's Comment*
19. *Five out of four people have trouble with fractions.*
20. *It's a rare person who wants to hear what he doesn't want to hear. – Dick Cavett*
21. *ime travel. Quantum experiment proves t*
22. *If you try to fail, and succeed, which have you done?*

23. *The kisses of an enemy are deceitful, but the wounds of a friend are faithful – Old Testament*
24. *You always find it in the last place you look. – pessimist*
Well, I sure hope you stop looking once you've found it. – wit
25. *Anomalies don't promise understanding; for that we need paradox.*
26. *Seek out paradox – that place where established principles collide in contradiction, where logic self-destructs.*
27. *Today I invented a new word – plagiarism.*
28. *I just sold my homing pigeon on eBay for the 13th time.*
- 29.
30. *The past is a different country; they do things differently there.*
31. *Problems worthy of attack, prove their worth by hitting back. – Gooks*
32. *An egotist is someone who is more interested in themselves than they are in me.*
33. *You always find it in the last place you look. – pessimist*
Well, I sure hope you stop looking once you've found it. – wit

Most people stop looking once they've found it – wit
Numbskull! Once it's found, stop looking. – wit

Curiosity arrives late, a little frustrated having lost her keys. She echoes the quote.
Most people do stop looking once they've found it. – Understanding
Quick, hide it! – Reason (find)
Once you find it, don't move it! – Proof (place)
Next time, look in the last place first. – Logic (last)
Don't look for it, feel for it, trust the 'force' little padawan. – Reason (look)
Optimists keep looking, there might be two! – Theory (the)
Duplicate it somewhere else, now it won't just be in the last place you look. – Experiment (it)
Exercise a little discipline, always look in ten places. – Mathematics (always)
Then there are those who don't lose it in the first place. – Dogma (in)
Lengthening silence, one by one, they all turn to look at Paradigm. Sheeze.
Next time, hide yourself, make it look for you. – Paradigm
"Thanks boys, I'll remember that."

34. *A mind once expanded by a new idea, never regains its original shape – Mark Twain*
- 35.
36. *FTL - Who misplaced reality? It was here a moment ago.*
37. *next*
- 38.

In Reserve

The English spelling rule, 'i' before 'e' except after 'c' disproved by science.
I am lying.

There are rules to creativity. I know because I just made them up.
There are rules to creativity: I just invented them.

Permutations - ePmrtutaoisn

Anyone who is not shocked by quantum theory has not understood it. – Niels Bohr.
is missing the first part. This sentence

*Let's debate free will – which position would you like to argue?
CanYouFixTheSpaceBarOnMyKeyboard? – E-mail to tech support
It's never too late to invent a time machine.
The essence of time travel is the childhood wish, "Make it didn't happen'." – Niven
We need a third type of causality. – Gisin*

*The once was a cowboy named bill,
Who loved long debates on free will.
With a paradox wheel,
He brought nature to heel,
And reduced 6 billion seconds to nil.*

Discoveries driven by paradox and principle are more useful than those arrived at through experiment. – Amanda Geffer

Understanding nature's principles are just that – understanding. It's what science is all about – Amanda Geffer

Seek out paradox – that place where established principles collide in contradiction, where logic self-destructs. – Amanda Geffer

The supreme task of the physicist is to arrive at those universal elementary laws from which the cosmos can be built up by pure deduction. – Einstein

Anomalies don't promise understanding; for that we need paradox. – Amanda Geffer

Figuring out which <assumptions> to abandon will earn physicists profound insights into the workings of the universe. – Amanda Geffer

If a picture is worth a thousand words, then a story is worth a thousand pictures, but a paradigm is worth a thousand stories.

The inventor of autocorrect just died. The funnel will be healed tomato.

Buy one burger for the price of two and get a second one absolutely free!

Parking available in empty spaces only.

I tried to remarry my ex, but she figured out I was only after my money.

ATTENTION. Oh, sorry, never mind.

The difference between science fiction and reality is about a generation.

*FIRST RULE OF PROOFREADING
Pay attention to detail.*

next

Comics

In Use

1. **Unknown:** “Wagon on Square Wheels” (need permission)
2. **Fox Trot:** “Logic” (need permission)
3. **Frank & Ernest:** “Counter-Intuitive”
4. **xkcd:** “Amber Particles of Grain” (with permission)
5. **Wilkinson:** “Estrogen Studies” (need permission)
6. **Foxtrot:** “Why do I always feel like I’m being watched?”
7. **Speed Bump:** “Prisoner Plagiarism” (need permission)
8. **Frank & Ernest:** “Global” (need permission)
9. **Frank & Ernest:** “Procrastination” (need permission)
10. **Frank & Ernest:** “Kleptomania clinic” (need permission)
11. **B.C.:** “Boomerang” (need permission)
12. **Luann:** “Horoscope” (need permission)
13. **Fox Trot:** “Football Pattern” (need permission)
14. **Ziggy:** “Hard to Think Outside the Box” (need permission)
15. **Far Side:** “My brain is full” (need permission)
16. **Escher:** “Drawing Hands” (need permission)
17. **Rose is Rose:** “Deceptive Checkup” (need permission)
18. **Shoe:** “Conundrum” (need permission)
19. **xkcd:** “Analogies” (need permission)
20. **Frazz:** “Hypothetically speaking” (need permission)
21. **Frank & Ernest:** “Inexperienced loan officer” (need permission)
22. **Mr. Spot:** “Borrow from the Future” (need permission)
23. **Far Side:** “Beach Boards” (need permission)
24. **Mother Goose and Grimm:** “God Sneezed” (need permission)
25. **Artist:** “Title”
26. **Artist:** “Title”
27. **Frank & Ernest:** “Zeno” (need permission)
28. **Artist:** “Title”

In Reserve

PhD in Liatology

Escher stair cases

BC Ground Hog

Potential Chapters

Text

- | | |
|---|--------------------------|
| 1. The Measurement Problem | 5. Classical vs. Quantum |
| 2. Nonlocality | 6. STR |
| 3. Interpretations – 8 Schools of Thought | 7. QT3 |
| 4. U vs. R | 8. SSI |
| | 9. NLL |

- | | |
|---|---|
| 10. Fast Hierarchy | 38. Laws of Form |
| 11. Fast Calculus | 39. Imaginary Truthvalues |
| 12. Temporal Paradox | 40. Digital Circuits |
| 13. Attractor Structures | 41. Sequential vs. Combinatorial |
| 14. Successor Functions | 42. Complex Truthvalues |
| 15. Propositional Calculus | 43. The Problem of Communicating
Paradigm Shifts |
| 16. EPR | 44. SLC |
| 17. Hilbert Spaces | 45. BTC |
| 18. Complex Numbers | 46. The Classical Ensemble |
| 19. Quantum Checkers | 47. Everett's Many Worlds
Interpretation |
| 20. Quantum Billiards | 48. PKC |
| 21. 3D Chess | 49. NPL |
| 22. Survivor | 50. The Prisoner's Dilemma |
| 23. 12-lsec thought Experiment | 51. Self-Collapse & Basis Selection |
| 24. Time Travel Stories – Indeterminism | 52. The Dual EPR Protocol |
| 25. VRP | 53. ASPDC |
| 26. NTR | 54. How Thin is the Present? |
| 27. The Issue of God | 55. Superposition – Entanglement –
Collapse |
| 28. Miracles | 56. Spooky Coins & Magic Envelopes |
| 29. Evolution | 57. Starbucks entanglement |
| 30. Free Will vs. Determinism | 58. Limit theorems |
| 31. Randomness & Optimal Play | 59. Free Will |
| 32. What is the Subject Matter of
Science? | 60. Conditional Temporal Paradox |
| 33. A Discontinuous Change in Strategy | 61. Negative Time Reconnaissance
(NTR) |
| 34. Temporal Interference, Using
Paradox to Sculpt Reality | 62. Creative Productivity |
| 35. The Excursion to Infinity | |
| 36. Cantor's Diagonal Argument | |
| 37. Gödel's Incompleteness Theorem | |

Text

Todo

Discourses and Chapters with missing elements.

1. Ch-1: Paradigms - missing spiral solution to the 9 dot puzzle.
2. Ch-7: Entanglement – paper references for Surez & Scarini.
3. Ch-8: QTP – refers to common spacelike causes, deleted from chapter 3.

QM Basics

Text

Discrete versus Continuous

State

Superposition

Measurement Bases

Conjugate Bases

Action

The Uncertainty Principle

Interference

Mixtures versus Ensembles

Delayed Choice

Separable States

Entangled States

Conflict with Relativity

text

Notes for the Future...

Text

Random Guess Problem

Correct attack is to change the problem from one discrete problem into a nearly smooth set of problems, and then analyze the complete set. As an aid to triaging why the original choice of percentages is paradoxical, we'll allow that any answer within 1% is good enough to be considered correct, think significant digits.

The first member of the set has all four choices at 0%. Then let choice D) increase in 1% increments up to 100%. Then reset D) to 0% and increment C) to 1%, and let D) scan up to 100% again. Repeat for each choice in turn, it's like an odometer. Note that we now have a set of more than 100 million problems (101^4). For most of these, there are no solutions. For any problem where one choice is 25% and the rest are different from each other, there is one right answer, which is whatever option has the 25%. Now if two options have 50% and the others differ from each other, then there are two correct answers, either of the options that list 50%. Similar arguments apply for 75% (3 right answers), and all 100% (4 right answers), and the last problem in the set.

What makes the original problem so compelling is the paradoxical nature of the answer. Most of the problems have no answers, some have real answers (1 to 4) and a few, like the original problem, have imaginary answers.

Boolean logic is incomplete. It is not closed under self-reference. Once self-reference is allowed into a formal system (or more typically cannot be prohibited), logic must be extended by adding imaginary truthvalues in order to restore the mathematical property of closure. Since all the problems in the set had only real answers, an imaginary answer is not one of the options. This allows us to tag such problems as having no solutions.

We are left however with the feeling (hunch, intuition) that the story is incomplete. So challenge, can closure in logical systems that admit self-reference be maintained by adding imaginary truthvalues (the extension to complex truthvalues is regarded as anticipated and necessary). Can all meta-paradoxes (an attempt to create a paradox on imaginary or complex truthvalues) be solved within the bounds of complex truthvalues? Is closure a chimera?

A multiple choice problem about the odds of guessing the correct solution to that very same multiple choice problem has garnered much thought and interest on the internet. In the original problem the odds were carefully chosen to make the answer of this self-referential problem paradoxical. The resolution presented here is an example of a problem solving technique called "change the problem." The resulting analysis suggests the need for imaginary truthvalues, an idea first proposed by G. Spencer-Brown in his 1969 book "Laws of Form."

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<https://www.linkedin.com/pulse/paradoxical-random-guess-problem-allan-goff>

Classical to Quantum Promotion

Question: Why do we only see the transition from classical states to quantum states? Why do we never see the transition from classical states to quantum states?

Answer: Perhaps self-reference has the power to transition an otherwise classical state into a quantum state.

Explanation: If a Boolean value is inverted, I've transformed one classical state into another. If an imaginary value is buffered, I've transformed one classical state into another. But if an inverter (NOT gate) or a buffer (BUF gate) feeds its output back into its input, the expected classical value has just gone indeterminate. It's been promoted from a single classical value into a superposition of classical values – isn't that the very definition of a quantum value?

Unpredictable

PD N:

Paradigm bounded in to be politely greeted by <character 1>, "How are you?"

"Unpredictable." Paradigm responded.

"What, oh say that's not bad, I might use it."

PD N+1:

Paradigm entered the room. <character 2> inquired, "How are you tonight?"

"Unpredictable. How about you?" Curiosity noted the exchange with little more than an eyebrow raise.

<character>, "I'm fine, I think. That's a good one."

PD N+2:

<character 3> "Paradigm, let me introduce <visitor>."

<visitor>, “And how are you?”

“Unpredictable.” Paradigm said with a grin.

“Ah...so that’s the kind of group you’ve been spending time with.”

Curiosity watched the whole exchange this time, and started formulated plans...

PD N+3:

Curiosity skipped over to Paradigm, dragging their mentor with her. “You’re a little late.” Then slyly looking at Understanding while addressing Paradigm, “How are you?”

Paradigm looked at her, hesitated for a moment to draw out the tension thus pretending to suggest that he was playing along, but said instead, “Irresistible.”

Curiosity did a double take. After recovering her wits, “That is not what I expected you to say.”

“What did you expect me to say?”

Curiosity, “What you usually say, ‘unpredictable’.”

Paradigm, leaning in, “So, was what I said predictable or unpredictable?”

“Aughh, get out of my head.”

Understanding, ruefully smiling but also shaking his head, “You keep finding new places where self-reference intrudes itself.”

[QTP](#)

Quantum Temporal Paradox

Question The Paradigm

Quest Tempts Paradigm

[Egghead Fun](#)

Q: What does a bunch of eggheads do for fun?

A: Reinvent anything – user ergonomics, spellings, architecture, ghost stories.

The 3-stroke engine.

The solar panel.

Space launch systems.

It’s called “What If Knite”

Is Paradigm late?

Deconstruction?

Dinner.

Cards.

Games.

Cleaning up envelopes.

What do they do in the Realm for fun?

How are they forwarding the quest in the meantime?

Are they doing literature searches?

Preparing papers, or lectures?

Some do good activity perhaps.

Do they participate in a charity?

A chess tournament.

Are they watching TV?

At a ball game?

Red Herrings?

Riddles?

Team building – Understanding's leadership skills?

Recreational reentry.

Sci fi convention.

Magic?

Door to door selling.

Cooking.

Graffiti?

Propose alternative spellings.

Reinvent anything.

TLA

Extend Reason's Uncertainty box.

- Touch
- Light
- Audio

Sum Over Histories

The classical ensemble is inherently linear, the superposition of classical realities.

Uneven weights can be handled by multiple identical copies?

Are multiple *roots* allowed?

Prunes and collapses occur when the classical ensemble fails to be self-consistent.

In QT3 pruning happened when a single reality was inconsistent with itself.

Each particle has a well-defined state vector.

- The bases are arbitrary, GS decomposition
-

What can be in superposition?

Location?

Existence nonexistence? Captured/not captured?

Phase angles? d/dx of the phase angle?

Action cube.

If inconsistent in one basis, inconsistent in all?

If reality $A \Rightarrow$ particle 1 in state z & reality $B \Rightarrow$ state y , then particle is half z & half y .

However, if reality $C \Rightarrow$ particle 1 has zero vector, then reality C is inconsistent.

If particle is half in state $-z$ & half in state z , is that the zero vector? Seems unlikely.

Given bases A, B, C .

Let particles x, y, z, w be in various superpositions of p -states, represented in each basis.

The system is inconsistent whenever there is no state vector that can represent all 3 superpositions.

Each reality evolves deterministically.

What about attractors.

There are trajectories that span attractors – sudden change in *state*.

So many ideas...

The cancellation of adjacent contradictions leads to collapse of excited states.

What does NLL look like in the classical ensemble?

Indistinguishable Quantum Tic-Tac-Toe (IQT versus DQT).

The original quantum tic tac toe game (dare I say *classic quantum tic tac toe*) made the hidden assumption that the moves of each player were distinguishable. In quantum physics indistinguishable particles have a big impact on Feynman's sum over histories calculation.

In distinguishable quantum tic-tac-toe (DQT), the history of the classical ensemble is a pure tree; which originally looked like a subset of the TTT game tree, and one that gets pruned as play progresses. In IQT, the ensemble history turns into a directed graph (unidirectional), but it allows for cycles; it's no longer a pure tree. Graph theory is a mature discipline in computer science, so the math has already been developed for us. Pruning links and nodes have the potential to yield a variety of quantum-like behaviors; including decoherence, interference, measurement, and even Everett/Wheeler's many-worlds interpretation.

On second thought, the game tree for classic tic-tac-toe (TTT) is also a directed graph. The 3 move gambit 1, 5, 9 yields the same position as the 9, 5, 1 gambit; O in the center square with X on opposite corners (don't even get me started on board rotation/reflection symmetries).

Here are the new thoughts; are there new forms of pruning (in addition to contradiction & collapse), and does pruning of directed graphs yield situations that can be mapped to major quantum phenomenon? The graph starts connected, not disjointed. Sufficient pruning could yield a disjointed graph – Everett/Wheeler's many-worlds interpretation or measurement? Let quantum checkers allow phase oscillation on every move. Then a jump can get to a square with a different phase angle than a regular two step move. Does that position (node) get pruned by destructive interference?

Omg – can't support multiple bases without some kind of phase angle; $|D\rangle = 1/\sqrt{2}\{|H\rangle + |V\rangle\}$, $|S\rangle = 1/\sqrt{2}\{|H\rangle - |V\rangle\}$.

I want to draw a distinction between contradictions within a single reality (DQT pruning) and ensemble contradictions (such as the QTP hypothesis or cyclic entanglement collapse in DQT which while it involves multiple realities doesn't seem to be a cycle in a graph). I'm expecting QTP pruning to involve a graph cycle.

I think you can see where this line of thought leads to.

Our present is defined by pasts which *aborted*, by *multiple* pasts which actually happened, and the possibility of both from *future* events as well. Where did Reality go? It was here a moment ago.

Planning

Text

Word count analysis

GEB AT – 400 words/page.

GEB text – 450 words/page.

PD – 600 - 700 words/page.

Ch – 650 words/page.

750 pages => $(440 / 650) * 750 = 508$ pages
 750 pages * 440 words/page = 330,000 words.

At 500 pages (Times, 12 pt) ~330,000 words, I'll be as complicated as GEB.

Jargon Questions

world view versus worldview
 truthvalues versus truth values

Requirements for FTL

1. Spacelike cause
 - Entanglement
 - Restriction rule
2. Determine left from right (SSI)
3. RC – Two spacelike causes with opposite Poincare' rotations
4. Avoiding temporal paradox
 - Packet based FTL where one bit is a random correlation.
 - Self-collapse in the indeterminate basis
5. Falsifiability – all the above.

Text

Other Characters

Did Paradigm ever date Epiphany? Yes.
 One of the nights when he's absent.
 Yep, and Curiosity finds out about it.
 But it takes a while to figure out who she is.
 Turns out she will be the impetus behind 3D Chess.

Goosey Apples

Curiosity overhears Mathematics etc. talking about goosey apples, she internalizes this as Carmel Apples, but the boys get it, and try not tease her.
 Her blond moments are few, but they do sneak up on her and she so, so hates the stereotype.

Curiosity entered the back room. Proof looked up and politely quired, "What are the others doing?"

"They're all engrossed in the new computation engine and munching, I think, on Carmel apples."

Logic, "There are Carmel apples? No one told me." He was about to leave the room, but Paradigm stopped him, "There are no Carmel apples here tonight." Looking at Curiosity, "Why do you think there are?"

Curiosity, "Look, I was hustling to join you all, so I only overheard a few words, something about 'goosey apples'. I just assumed..."

Theory, Logic and Proof; their eyes widened in unison, their mouths opened, then closed and in quick succession each bit their lips trying to keep quiet. Paradigm noticed and shook his head. Curiosity was one of the smartest people he knew, but from time to time, blond moments seemed to take special interest in her and she so, so hated the stereotype. He redirected the conversation back to 3D chess, compassion trumping his love of wit.

He almost pulled it off, but Curiosity caught on.

Hands on hips, Curiosity glared at her comrades, “I don’t know what you aren’t telling me, ‘gentlemen’, but when I figure it out, there’s going to be hell to pay.” Paradigm, diplomatically, made a last if feeble effort to guide them back to the task at hand. With long suffering graciousness, Curiosity acquiesced, “As you wish.”

Tool Notes

From Word in the new Office 365 - Test of filename auto field. Can’t get filename to change from template nor word count to change from 12. Not enough to print preview and save. Must actually have text changes to save.

Publishing

Robert Brekke, is in my phone, VSPCA

Science Fact publishers

Bantam Doubleday Dell

Johns Hopkins University Press

University of California Press

Simon & Schuster

Farrar, Straus & Giroux’s North Point Press

Harvard University Press

Princeton University Press

Cliff Street Books

Sarah Jane Freymann Literary Agency

W. W. Norton

Commercial or trade press

Competitive Analysis

1. Jim Al-Khalili, *Paradox*
 - a. Ch-7 The Grandfather Paradox – adequate presentation
2. Nick Herbert, *Quantum Reality*
 - a. 8 interpretations

Blurb

Physicists have been reluctant to tackle the foundations of quantum mechanics because none of the mental models suggested by the math are correct. Over and over again, popular writers have attempted to make some sense of this but without success. The earliest summary of the challenges was compiled by Nick Herbert in his 1985 book *Quantum Reality*. In chapters 9 & 10 he lays out the 8 leading interpretations of quantum mechanics; none make any different experimental predictions.

Since then, dozens of popular level books have been written striving to make sense of quantum mechanics. These bold authors were not content to ‘shut up and compute,’ they wanted to *understand* and had the courage to take a stab at it. Their efforts range from the

- very technical; Roger Penrose, *The Emperor’s New Mind*, Roland Omnes, *The Interpretation of Quantum Mechanics*,
- to the popular; Brian Green, *The Fabric of the Cosmos*, David Deutsch, *The Fabric of Reality*,
- to the lament; Lee Smolin, *The Trouble with Physics*, Peter Woit, *Not Even Wrong*,
- to the educational; David Z Albert, *Quantum Mechanics and Experience*, R. I. G. Hughes, *The Structure and Interpretation of Quantum Mechanics*,
- to the desperate; Jeffery A. Barrett, *The Quantum Mechanics of Minds and Worlds*, Michael Talbot, *The Holographic Universe*,
- to its worse than we thought, Tim Maudlin, *Quantum Non-Locality and Relativity*.

In each of these books the authors lay out a foundation of the topic, introduce the conundrums, discuss various potential solutions, and universally come to no conclusions. Quantum mechanics (QM) is far and away our most successful physical theory, yet at a fundamental level we have failed to achieve any understanding of what it actually *means*. QM is thumbing its nose at us. It is, frankly, a professional embarrassment.

The difficulties do not stop at quantum mechanics.

- Even mathematics has not been spared the concern that all is not right with the scientific enterprise; Morris Kline, *Mathematics: The Loss of Certainty*, Robert Gilmore, *Catastrophe Theory (chapter 1)*,
- nor logic itself: G. Spencer-Brown, *Laws of Form*, N. S. Hellerstein, *Diamond; A Paradox Logic.*, Raymond M. Smullyan, *Diagonalization and Self-Reference*.

The one pillar of light, our one source of hope, remains Thomas S. Kuhn’s seminal work, *The Structure of Scientific Revolutions* in which he introduces the modern meaning of the word paradigm. The problem is not in nature, it is rather in our inability to rise above the world-views that constrain our thinking. For the first time since the inception of science, nature is outsmarting us.

PoP takes up this quest with two foundational perspectives, a new tool, and a unifying hypothesis. The first perspective is that 100 years into the quantum revolution the field is now ripe for a paradigm shift (Kuhn), the second perspective is that self-reference is the key to resolving these conundrums (G. Spencer-Brown). Together these two perspectives lead to a new tool, the invention of imaginary truthvalues (Hellerstein), a fundamental extension of symbolic logic which can handle self-reference at a formal level, one that proposes conjugate bases as part of the foundation of logic. This tool and its supporting perspectives then suggest the hypothesis of QTP – Quantum Temporal Paradox (Goff); that cyclic entanglement in time is the resolution of the quantum measurement problem. QTP resolves most of the remaining issues of understanding quantum physics.

It's a bold program, and until experiments are devised and run, we won't know if nature is this way, or if QM has once again thwarted our best efforts at achieving understanding. The purpose of Pop, in part, is to show where in the landscape such experiments may exist.

BookScan

Web site that reveals book sales? Pricy.

Christian - <https://www.ecpa.org/Login.aspx?returl=>

Consulting Editor – Alan Rinzler - <https://alanrinzler.com/services/>

Permissions

King Features

Go Comics – web site.

Todo

1. Look up Surez & Scarini references
2. Look up Japanese grad students on QT3 perfect play
3. Mark up text with [REF]
4. Fill in the REFs
5. Get permissions for the comics
6. Find a commercial artist

Next

Research

Next

EvenOdd Cyclic Entanglements

Just ran into some interesting difficulties in the QTP hypothesis. Three unary gates in a feedback loop are paradoxical in either the Boolean bases or the imaginary basis, and indeterminate in the other. However, four unary gates in a feedback loop are indeterminate in both, or paradoxical in both.

WTF?

Random Defeating Conjugate Bases FTL Protocol

Here's the problem. Creating a cyclic entanglement is different than measuring one. QT3 grows entanglements until they are cyclic, and that quite likely requires a nonlinear operator. Quantum Othello creates an EPR entanglement complete. Note that in QT3 EPR states collapse, but they do not in physics. Even if we can grow semi-entanglements into cyclic ones, any measurement prior to completion destroys it prematurely. If any measurement must be in the future light cones of every member of the entanglement, then FTL is off the table. Note that this is the case for the current production of EPR pairs.

So what we need is a 6 to 8 element entanglement with conjugate bases such that the correlations between the receiving elements depend on the measurement bases. The entanglement must be cyclic in both bases. Let's assume 8 elements, 4 for sending, 4 for receiving. The sender measures his four in one of two bases, so does the receiver. Label the 4 sending elements 1, 3, 7, 9 and the four receiving elements 2, 4, 6, 8. Then a duplex protocol has to look something like the following table.

Table: FTL protocol – duplex, high bandwidth between corners & sides via measurements in two conjugate bases

Sending	Receiving			
	C basis		D basis	
C basis	13~C	24~C	13~A	24~C
	79~C	68~C	79~C	68~A
D basis	13~C	24~A	13~A	24~A
	79~A	68~C	79~A	68~A

If we have to tolerate some random correlations, then bandwidth merely suffer, because bits would have to be encoded in a packet of entanglements instead of one bit/entanglement. May also turn out that simplex is best we can do. The key is a many particle cyclic entanglement, that can be in the past light cones of all the measurements, where each participant has two measurement choices, and a double pair of quantum objects to measure. Seems like there is enough degrees of freedom in an 8x8 matrix to allow at least one of these to be feasible.

Table 2: Ditto, but with 6 objects and packets

	Receiving				
Sending	C basis		D basis		
C basis	13~C	24~C	000 - 111	24~A	000-111
	17~C	6~RF			001-110
	37~C				
D basis		26~A		46~A	010-101
					011-100

GHZ

Best guess is that one can only prune half the states in both conjugate bases simultaneously. That would explain GHZ. A system cyclically entangled in one basis, will be maximally indeterminate in any conjugate basis. Therefore, consider this one

$$|H\rangle = \frac{1}{\sqrt{2}}\{|D\rangle + |S\rangle\}$$

$$|V\rangle = \frac{1}{\sqrt{2}}\{|D\rangle - |S\rangle\}$$

Each of the three particles is individually in a superposition of horizontal and vertical polarization

$$\begin{aligned}\Psi_1 &= \frac{1}{\sqrt{2}}\{|H\rangle_1 \pm |V\rangle_1\} \\ \Psi_2 &= \frac{1}{\sqrt{2}}\{|H\rangle_2 \pm |V\rangle_2\} \\ \Psi_3 &= \frac{1}{\sqrt{2}}\{|H\rangle_3 \pm |V\rangle_3\}\end{aligned}\tag{2}$$

Next

$$\Psi_{123} = \frac{1}{\sqrt{2}} \{ (|H\rangle_1)_c (|H\rangle_2 |V\rangle_3)_s \pm (|V\rangle_1)_c (|V\rangle_2 |H\rangle_3)_s \} \quad (2)$$

Where the parentheses indicate spatial separation; there is one particle (#1), labeled 'c' (corners) that is to be regarded as spacelike separated from the other two (#2, #3) labeled 's' (sides). Note that any measurement on any one of the three particles will reveal a perfectly random outcome. However, they are very particularly correlated. In particular, the two particles on the sides side of the universe are anti-correlated.

First assume that all measurements are in the horizontal basis. Assume that particle #1 is measured first. Then any subsequent measurement of particles #2 & #3 will show them to be anticorrelated. In fact, the order doesn't matter, they will always be seen to be anticorrelated.

Now let the measurement of particle #1 be in the diagonal basis. At first it appears that this collapses the other two particles also in the diagonal basis

$$\begin{aligned} \Psi_2 &= \frac{1}{\sqrt{2}} \{ |D\rangle_2 \pm |S\rangle_2 \} \\ \Psi_3 &= \frac{1}{\sqrt{2}} \{ |D\rangle_3 \pm |S\rangle_3 \} \end{aligned}$$

If so, then any subsequent measurement on these two particles in the horizontal basis will randomize them, and they will no longer be anti-correlated.

Looks like we finally have a measurement based FTL simplex protocol; but perhaps we should do the math. That means casting the twin state in the horizontal basis into the diagonal basis. A bunch of substitutions and multiplications are in order.

$$\begin{aligned} \Psi_{123} &= \frac{1}{\sqrt{2}} \{ (|H\rangle_1)_c (|H\rangle_2 |V\rangle_3)_s + (|V\rangle_1)_c (|V\rangle_2 |H\rangle_3)_s \} \quad (2) \\ \Psi_{123} &= \frac{1}{\sqrt{8}} \{ (|D\rangle_1 + |S\rangle_1)_c (|D\rangle_2 + |S\rangle_2) (|D\rangle_3 - |S\rangle_3)_s \\ &\quad + (|D\rangle_1 - |S\rangle_1)_c (|D\rangle_2 - |S\rangle_2) (|D\rangle_3 + |S\rangle_3)_s \} \\ \Psi_{123} &= \frac{1}{\sqrt{8}} \left\{ \begin{aligned} &+|D\rangle_1 |D\rangle_2 |D\rangle_3 - |D\rangle_1 |D\rangle_2 |S\rangle_3 + |D\rangle_1 |S\rangle_2 |D\rangle_3 - |D\rangle_1 |S\rangle_2 |S\rangle_3 \\ &+|S\rangle_1 |D\rangle_2 |D\rangle_3 - |S\rangle_1 |D\rangle_2 |S\rangle_3 + |S\rangle_1 |S\rangle_2 |D\rangle_3 - |S\rangle_1 |S\rangle_2 |S\rangle_3 \\ &+|D\rangle_1 |D\rangle_2 |D\rangle_3 + |D\rangle_1 |D\rangle_2 |S\rangle_3 - |D\rangle_1 |S\rangle_2 |D\rangle_3 - |D\rangle_1 |S\rangle_2 |S\rangle_3 \\ &-|S\rangle_1 |D\rangle_2 |D\rangle_3 - |S\rangle_1 |D\rangle_2 |S\rangle_3 + |S\rangle_1 |S\rangle_2 |D\rangle_3 + |S\rangle_1 |S\rangle_2 |S\rangle_3 \end{aligned} \right\} \\ \Psi_{123} &= \frac{1}{\sqrt{8}} \left\{ \begin{aligned} &+|D\rangle_1 |D\rangle_2 |D\rangle_3 - |D\rangle_1 |S\rangle_2 |S\rangle_3 \\ &-|S\rangle_1 |D\rangle_2 |S\rangle_3 + |S\rangle_1 |S\rangle_2 |D\rangle_3 \\ &+|D\rangle_1 |D\rangle_2 |D\rangle_3 - |D\rangle_1 |S\rangle_2 |S\rangle_3 \\ &-|S\rangle_1 |D\rangle_2 |S\rangle_3 + |S\rangle_1 |S\rangle_2 |D\rangle_3 \end{aligned} \right\} \end{aligned}$$

$$\Psi_{123} = \frac{1}{\sqrt{4}} \left\{ \begin{array}{l} +|D\rangle_1|D\rangle_2|D\rangle_3 - |D\rangle_1|S\rangle_2|S\rangle_3 \\ -|S\rangle_1|D\rangle_2|S\rangle_3 + |S\rangle_1|S\rangle_2|D\rangle_3 \end{array} \right\}$$

Ok, now with the minus sign

$$\begin{aligned} \Psi_{123} &= \frac{1}{\sqrt{2}} \{ (|H\rangle_1)_c (|H\rangle_2|V\rangle_3)_s - (|V\rangle_1)_c (|V\rangle_2|H\rangle_3)_s \} \\ \Psi_{123} &= \frac{1}{\sqrt{8}} \{ (|D\rangle_1 + |S\rangle_1)_c (|D\rangle_2 + |S\rangle_2) (|D\rangle_3 - |S\rangle_3)_s \} \\ \Psi_{123} &= \frac{1}{\sqrt{8}} \left\{ \begin{array}{l} +|D\rangle_1|D\rangle_2|D\rangle_3 - |D\rangle_1|D\rangle_2|S\rangle_3 + |D\rangle_1|S\rangle_2|D\rangle_3 - |D\rangle_1|S\rangle_2|S\rangle_3 \\ +|S\rangle_1|D\rangle_2|D\rangle_3 - |S\rangle_1|D\rangle_2|S\rangle_3 + |S\rangle_1|S\rangle_2|D\rangle_3 - |S\rangle_1|S\rangle_2|S\rangle_3 \\ -|D\rangle_1|D\rangle_2|D\rangle_3 - |D\rangle_1|D\rangle_2|S\rangle_3 + |D\rangle_1|S\rangle_2|D\rangle_3 + |D\rangle_1|S\rangle_2|S\rangle_3 \\ +|S\rangle_1|D\rangle_2|D\rangle_3 + |S\rangle_1|D\rangle_2|S\rangle_3 - |S\rangle_1|S\rangle_2|D\rangle_3 - |S\rangle_1|S\rangle_2|S\rangle_3 \end{array} \right\} \\ \Psi_{123} &= \frac{1}{\sqrt{8}} \left\{ \begin{array}{l} -|D\rangle_1|D\rangle_2|S\rangle_3 + |D\rangle_1|S\rangle_2|D\rangle_3 \\ +|S\rangle_1|D\rangle_2|D\rangle_3 - |S\rangle_1|S\rangle_2|S\rangle_3 \\ -|D\rangle_1|D\rangle_2|S\rangle_3 + |D\rangle_1|S\rangle_2|D\rangle_3 \\ +|S\rangle_1|D\rangle_2|D\rangle_3 - |S\rangle_1|S\rangle_2|S\rangle_3 \end{array} \right\} \\ \Psi_{123} &= \frac{1}{\sqrt{4}} \left\{ \begin{array}{l} -|D\rangle_1|D\rangle_2|S\rangle_3 + |D\rangle_1|S\rangle_2|D\rangle_3 \\ +|S\rangle_1|D\rangle_2|D\rangle_3 - |S\rangle_1|S\rangle_2|S\rangle_3 \end{array} \right\} \end{aligned} \quad (2)$$

That notation is tedious. And I want to double check phase mixtures.

Let's try this one.

The plus-plus permutation: all R, 23 half AC half CC

$$\begin{aligned} \Psi_{123}^H &= \frac{1}{\sqrt{2}} \{ +HHV + VVH \} \\ \Psi_{123}^H &= \frac{1}{\sqrt{2}} \{ + (H)(H)(V) + (V)(V)(H) \} \\ \Psi_{123}^D &= \frac{1}{\sqrt{16}} \{ + (D+S)(D+S)(D-S) + (D-S)(D-S)(D+S) \} \\ \Psi_{123}^D &= \frac{1}{\sqrt{16}} \{ +DDD - DDS + DSD - DSS + SDD - SDS + SSD - SSS \} \\ \Psi_{123}^D &= \frac{1}{\sqrt{8}} \{ +DDD - DSS - SDS + SSD \} \\ \Psi_{123}^D &= \frac{1}{\sqrt{4}} \{ +DDD - DSS - SDS + SSD \} \end{aligned}$$

The plus-minus permutation: all R, 23 half AC half CC

$$\Psi_{123}^H = \frac{1}{\sqrt{2}} \{ +HHV - VVH \}$$

$$\begin{aligned}
\Psi_{123}^D &= \frac{1}{\sqrt{16}} \{+(D+S)(D+S)(D-S) - (D-S)(D-S)(D+S)\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{16}} \{+DDD - DDS + DSD - DSS + SDD - SDS + SSD - SSS\} \\
&\quad - \frac{1}{\sqrt{16}} \{-DDD - DDS + DSD + DSS + SDD + SDS - SSD - SSS\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{8}} \{-DDS + DSD + SDD - SSS\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{4}} \{-DDS + DSD + SDD - SSS\}
\end{aligned}$$

The minus-plus permutation: all R, 23 half AC half CC

$$\begin{aligned}
\Psi_{123}^H &= \frac{1}{\sqrt{2}} \{-HHV + VVH\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{16}} \{-(D+S)(D+S)(D-S) + (D-S)(D-S)(D+S)\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{16}} \{-DDD + DDS - DSD + DSS - SDD + SDS - SSD + SSS\} \\
&\quad + \frac{1}{\sqrt{16}} \{+DDD + DDS - DSD - DSS - SDD - SDS + SSD + SSS\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{8}} \{+DDS - DSD - SDD + SSS\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{8}} \{+DDS - DSD - SDD + SSS\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{4}} \{+DDS - DSD - SDD + SSS\}
\end{aligned}$$

The minus-minus permutation: all R, 23 half AC half CC.

$$\begin{aligned}
\Psi_{123}^H &= \frac{1}{\sqrt{2}} \{-(H)(H)(V) - (V)(V)(H)\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{16}} \{-(D+S)(D+S)(D-S) - (D-S)(D-S)(D+S)\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{16}} \{-DDD + DDS - DSD + DSS - SDD + SDS - SSD + SSS\} \\
&\quad - \frac{1}{\sqrt{16}} \{-DDD - DDS + DSD + DSS + SDD + SDS - SSD - SSS\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{8}} \{-DDD + DSS + SDS - SSD\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{8}} \{-DDD + DSS + SDS - SSD\} \\
\Psi_{123}^D &= \frac{1}{\sqrt{4}} \{-DDD + DSS + SDS - SSD\}
\end{aligned}$$

So, my misunderstanding is that collapse selects just one term. That's not what it does, it merely divides. Thus, when particle 1 is measured we end up with either of two states still entangling particles 2 & 3. From this state

$$\Psi_{123}^D = \frac{1}{\sqrt{4}} \{DDD - DSS - SDS + SSD\}$$

We will divide and convert back to the horizontal basis for particle 1 being D

$$\begin{aligned}
\Psi_{123}^D &= \frac{1}{\sqrt{2}} \{(D)(D) - (S)(S)\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{4}} \{(H+V)(H+V) - (H-V)(H-V)\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{4}} \{HH + HV + VH + VV - HH + HV + VH - VV\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{4}} \{HV + VH + HV + VH\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{2}} \{HV + VH\}
\end{aligned}$$

And for particle 1 being S

$$\begin{aligned}
\Psi_{123}^D &= \frac{1}{\sqrt{2}} \{-(D)(S) + (S)(D)\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{4}} \{-(H+V)(H-V) + (H-V)(H+V)\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{4}} \{-HH + HV - VH + VV + HH + HV - VH - VV\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{4}} \{+HV - VH + HV - VH\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{2}} \{+HV - VH\}
\end{aligned}$$

Now the CC with 0 phase

$$\begin{aligned}
\Psi_{123}^D &= \frac{1}{\sqrt{2}} \{(D)(D) + (S)(S)\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{4}} \{(H+V)(H+V) + (H-V)(H-V)\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{4}} \{HH + HV + VH + VV + HH - HV - VH + VV\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{4}} \{HH + VV + HH + VV\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{2}} \{HH + VV\}
\end{aligned}$$

And for AC with 0 phase

$$\begin{aligned}
\Psi_{123}^D &= \frac{1}{\sqrt{2}} \{(D)(S) + (S)(D)\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{4}} \{(H+V)(H-V) + (H-V)(H+V)\} \\
\Psi_{123}^H &= \frac{1}{\sqrt{4}} \{+HH - HV + VH - VV + HH + HV - VH - VV\}
\end{aligned}$$

$$\Psi_{123}^H = \frac{1}{\sqrt{4}}\{+HH - VV + HH - VV\}$$

$$\Psi_{123}^H = \frac{1}{\sqrt{2}}\{+HH - VV\}$$

Bell States in Conjugate Bases

Would like to see this as a table

HV	DS
HH + VV	DD + SS
HH - VV	DS + SD
HV + VH	DD - SS
HV - VH	SD - DS

Decoherence

Stream of consciousness...

Two models of decoherence;

1. Extending entanglements ala QT3
2. Diluting into the larger phase space of the combined QC systems

Potentially degenerate.

Imagine a growing linear chain of semi-entanglements. Each quantum object in the chain is in a simple two-way superposition. Now imagine measuring every object. Those near the end of the chain are very nearly classical, those near the middle are very nearly in equally weighted superpositions.

Now turn the linear chain into a pure cyclic entanglement by connecting the endpoints; every object is now in a full equally weighted superposition. Repeat but now randomly connect any two objects. In general, there will be a central cyclic entanglement with two stems. The state of the objects in the central cycle are perfect superpositions, but the state of the objects in the stems are now classical, with the very values they were tending towards; they have been measured.

Random QT3

Consider a large tic-tac-toe board, absent winning criteria, say something on the size of a Go board (19x19 = 361). Add spooky-pairs randomly. The initial superpositions are isolated so retain their pure superposition status, and the number of entanglements initially tracks the number of superpositions. But at about the birthday point, new moves begin to couple existing superpositions creating chained entanglements.

As the size of the entanglements grow, the odds of extending each tip diminish while the odds of producing branched entanglements increase. The larger an entanglement the more likely the next move will extend it. Each extension of a tip, adds one element to the branched entanglement but no additional tips, making a new branch increasingly likely. Each new branch adds a tip, but no new member to the body, making a new branch less likely, so the number of tips to body elements is under negative feedback and should tend to 50%. The growth rate of an entanglement is proportional to its size so it's under positive feedback, winner take all.

Eventually a cyclic entanglement is created, but the larger the entanglement when it is 50/50 body and tips, the smaller the cycle and the larger the number of measured objects.

QT3 Model of Decoherence

So here is an interesting computer program. Grow until the first cyclic entanglement. Log the number of moves and entanglements, first extension, first cross coupling, average size & distribution (sigma, will be heavily skewed), the tip to body ratio (avg & sigma, will be tighter than gaussian), peak number of entanglements, the size of the collapsing entanglement, and its number of pure superpositions versus classical values and ratio. Do a number of runs, say 100 to get stats, and then repeat for different sized boards say 100, 200, 300, ..., 9,900, 10,000. That implies 100,000 runs. Then plot key stats versus board size.

Expect number of moves till first cycle to be about $\sqrt{N}$, where N is the number of squares on the board, and the ratio of superpositions to classical on the cyclic entanglement to be about $\log_2(N)$.

Columns:

1. Board size N : [100, 200, 300, ..., 9,900, 10,000]
2. Run size (100)
3. First extension ($\langle n \rangle \sigma$) (n)
4. First cross coupling ($\langle n \rangle \sigma$) (n) expect birthday answer
5. Max number of entanglements ($\langle n \rangle \sigma$) (e)
 - a. Save sequence to file for later plotting
6. First cyclic entanglement ($\langle n \rangle \sigma$) (n) expect σ to diverge
7. Size ($\langle n \rangle \sigma$) (# of nodes)
8. Cycle (superposition) count ($\langle n \rangle \sigma$) (# of nodes)
9. Collapse (classical) count ($\langle n \rangle \sigma$) (# of nodes)
10. Cycle to size ratio ($\langle n \rangle \sigma$)
11. Final number of entanglements ($\langle n \rangle \sigma$) (e)
12. Tip to size ratio for all the entanglements ($\langle n \rangle \sigma$) expect tight σ

This could be a scientific paper. Conclusion is how does this model of decoherence compare with others? Is it a solution to the measurement problem?

References

next

[What Should We Be Worried About?](#)

Next

146 - Unfriendly Physics, Monsters from the Id, and Self-Organizing Collective Delusions

168 - Magic

180 - No Surprises from the LHC: No Worries for Theoretical Physics

265 - Excellence

next